Sharp entropy retention under edge deletion in roses and parallel-edge graphs

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Abstract

Let a finite graph have strictly positive edge lengths, and let its entropy be the exponential growth rate of reduced circuits per unit metric length. We determine the smallest possible fraction of entropy retained by an optimal one-edge deletion for two graph families. For a rose with $r \geq 3$ loops, this fraction is $\log(2r-3)/\log(2r-1)$. For two vertices joined by $k \geq 4$ parallel edges, it is $\log(k-2)/\log(k-1)$. In each family, deleting any longest edge is optimal, and equality in the sharp bound holds exactly when all edge lengths are equal. The rose result proves the sharp three-loop bound and the uniform lower bound conjectured by Hamed, Aougab and Clay. The proof uses scalar Perron equations for entropy, a change of variables to a concave logistic-power function, and concentration of coordinates under a fixed-sum constraint. Repeated deletion gives sharp bounds for retaining several edges.

1 Introduction

Deleting an edge restricts the available reduced circuits of a metric graph and cannot increase their exponential growth rate. The entropy loss depends both on the combinatorial graph and on its edge lengths. The subgraph-entropy problem asks how much entropy can always be retained in a proper subgraph when that subgraph inherits the original metric.

Hamed, Aougab and Clay proved a positive rank-dependent lower bound for this problem and obtained stronger estimates for roses [1]. Following their Proposition 4.2, they conjectured that the sharp three-loop rose constant is $\log 3/\log 5$ and that no rose of higher rank has a smaller optimal retained fraction. We prove both assertions and determine the exact constant in every rank. We also solve the corresponding optimization problem for two vertices joined by parallel edges. The four-edge member of this second family is the example proposed in [1] for the general extremum.

Write R_r for the rose with r loop edges, and B_k for the graph consisting of two vertices and k parallel edges. Thus B_k has fundamental-group rank $k-1$. A metric ℓ assigns a strictly positive length to every unoriented edge. We write $h(G,\ell)$ for its metric entropy and $G-e$ for deletion of the unoriented edge e, with all other lengths unchanged. Define

$$D(G,\ell) = \max_{e \in E(G)} \frac{h(G-e, \ell|_{G-e})}{h(G,\ell)}. \tag{1.1}$$

Throughout the paper, logarithms are natural. The parent and the retained subgraph are never normalized separately.

Theorem 1.1. For every strictly positive metric on R_r, where $r \geq 3$,

$$D(R_r, \ell) \geq \frac{\log(2r-3)}{\log(2r-1)}. \tag{1.2}$$

For every strictly positive metric on B_k, where $k \geq 4$,

$$D(B_k, \ell) \geq \frac{\log(k-2)}{\log(k-1)}. \tag{1.3}$$

In either family, the edges attaining the maximum in (1.1) are exactly the longest edges. Equality in (1.2), respectively (1.3), holds if and only if all edge lengths are equal.

Equal lengths show that the constants cannot be increased. Their role in the theorem is stronger than providing examples: they are the unique minimizing metrics up to common scale. The assertions cover all positive length vectors, without restrictions on the ratios between their entries.

Figure 1 shows the smallest cases covered by the theorem at their minimizing metrics.

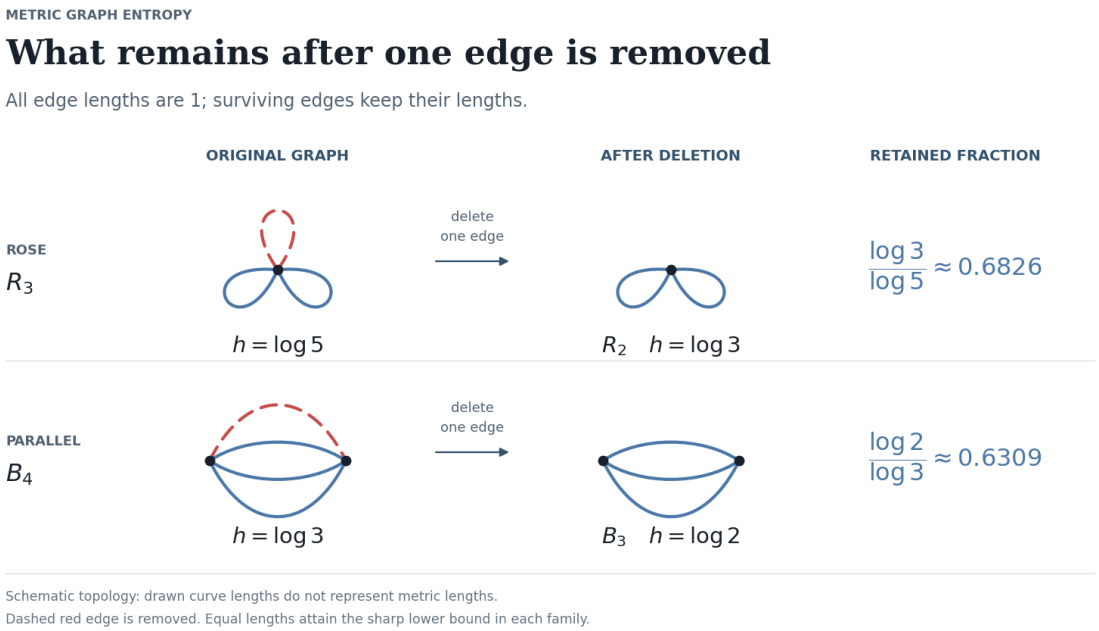


Figure 1: Unit-length examples attaining the sharp bounds: deleting one loop from R_3 retains the fraction $\log 3/\log 5$, and deleting one edge from B_4 retains $\log 2/\log 3$. The dashed red edge is removed; all surviving edges keep their original metric lengths. The drawings specify topology, not metric scale, and every edge is an optimal deletion in these equal-length examples.

For comparison with the rose estimate in [1], normalize the parent entropy to one, put $a = \log(2r-3)$, and let L be the second-longest loop length. After a longest loop is deleted, equation (4.3) of [1] gives $h' \geq 1 - 4/L$. Since all retained lengths are at most L, comparison with the equal-length rose also gives $h' \geq a/L$. The two estimates therefore imply

$$h' \geq \max\{1 - 4/L, a/L\} \geq \frac{a}{a+4}. \tag{1.4}$$

This optimized consequence of the earlier argument is still below (1.2). Its deficit from one is of order $1/\log r$, whereas

$$1 - \frac{\log(2r-3)}{\log(2r-1)} = \frac{\log(1 + 2/(2r-3))}{\log(2r-1)} \sim \frac{1}{r\log(2r)}. \tag{1.5}$$

The scalar entropy equation for roses used below is established; in particular, it appears as Lemma 5 of Balacheff and Merlin [2]. We derive it from the oriented-edge matrix to fix the conventions and obtain the parallel-edge equation in the same notation. Fixed-volume entropy minimization, as studied by Lim [3], concerns a different quotient: lower bounds for the parent and child entropies at their respective volumes do not themselves provide a lower bound for their ratio. The edge-addition formulas of Kim and Lim [4] describe entropy through path generating functions. Here the remaining task after the scalar reduction is a uniform comparison between the parent and deletion roots.

That comparison becomes an elementary concavity problem. After normalizing the parent entropy to one, we set $p_i = (1 + e^{\ell_i})^{-1}$. The entropy equation fixes the sum of these coordinates, and testing a smaller entropy replaces p_i by

$$\Phi_s(p_i) = \frac{p_i^s}{p_i^s + (1 - p_i)^s}, \qquad 0 < s < 1.$$

This function is strictly concave on $[0, 1/2]$. Once a minimum coordinate has been deleted, concentrating the remaining mass gives a one-variable lower bound. For roses there is one branch; for parallel-edge graphs the upper bound $p_i \le 1/2$ creates two branches. Their endpoint values give the constants and the equality cases.

2 Entropy equations and optimal deletion

A reduced circuit is a closed oriented edge path with no immediate reversal, including between its last and first edges. Entropy is the exponential growth rate, per unit metric length, of such circuits of length at most T as T tends to infinity. Whether one distinguishes a marked starting edge or identifies cyclic shifts does not change this rate. For the graphs considered here, it is characterized by the weighted non-backtracking matrix [1, §2.2].

Let $\vec{E}(G)$ be the oriented edges, with reversal $e \mapsto \bar{e}$ and initial and terminal vertices $o(e), t(e)$. Give both orientations the same length. For $s \ge 0$, set

$$M_G(s)_{ef} = \mathbf{1}_{\{t(e)=o(f),\, f \ne \bar{e}\}} e^{-s\ell(f)}. \tag{2.1}$$

For R_r with $r \ge 2$ and B_k with $k \ge 3$, this matrix is irreducible. At $s = 0$ its spectral radius is $2r - 1$ or $k - 1$, respectively; as s increases, the spectral radius decreases strictly to zero. Consequently

$$\rho(M_G(h(G, \ell))) = 1 \tag{2.2}$$

determines a unique positive entropy. The same facts apply to every deletion appearing in Theorem 1.1. They also give the scaling identity

$$h(G, c\ell) = c^{-1} h(G, \ell), \qquad c > 0. \tag{2.3}$$

Lemma 2.1. The entropy equations for the two families are

$$\sum_{i=1}^{r} \frac{1}{1 + e^{h\ell_i}} = \frac{1}{2} \quad \text{for } R_r, \qquad \sum_{i=1}^{k} \frac{1}{1 + e^{h\ell_i}} = 1 \quad \text{for } B_k. \tag{2.4}$$

Proof. Put $x_i = e^{-h\ell_i}$. For a rose, reversing every oriented edge commutes with $M_{R_r}(h)$. Uniqueness of the normalized positive Perron vector therefore makes its two coordinates for loop i equal; write their common value as v_i. At eigenvalue one,

$$v_i = 2 \sum_j x_j v_j - x_i v_i.$$

With $S = 2\sum_j x_j v_j > 0$, this gives $v_i = S/(1 + x_i)$. Substitution in the definition of S yields $\sum_i x_i/(1 + x_i) = 1/2$, which is the first equation in (2.4).

For B_k, orient each edge from the first vertex to the second and list the reverse orientations afterward. The matrix has block form

$$M_{B_k}(h) = \begin{pmatrix} 0 & C \\ C & 0 \end{pmatrix}, \qquad C_{ij} = \begin{cases} x_j, & i \neq j, \\ 0, & i = j. \end{cases} \tag{2.5}$$

Swapping the two orientation classes preserves the matrix, so the positive Perron vector is (v, v). Its equation is

$$v_i = \sum_j x_j v_j - x_i v_i.$$

Now $S = \sum_j x_j v_j$ gives $\sum_i x_i/(1 + x_i) = 1$, proving the second equation. The period two of (2.5) does not affect the positive Perron vector or introduce a factor of two into h.

Conversely, either scalar equation makes the vector with coordinates $v_i = (1 + x_i)^{-1}$, repeated in the two orientations, a strictly positive eigenvector at eigenvalue one. Irreducibility then gives (2.2). Thus the scalar equations characterize the entropy. □

For equal edge lengths L, Lemma 2.1 gives

$$h(R_r, L) = \frac{\log(2r - 1)}{L}, \qquad h(B_k, L) = \frac{\log(k - 1)}{L}. \tag{2.6}$$

It also orders the possible deletions.

Lemma 2.2. For R_r with $r \geq 3$ and B_k with $k \geq 4$, deletion of an edge of length ℓ_i retains at least as much entropy as deletion of an edge of length ℓ_j if and only if $\ell_i \geq \ell_j$. The entropy comparison is strict when the length comparison is strict.

Proof. Write $A = 1/2$ for roses and $A = 1$ for parallel-edge graphs. The entropy h_i after deleting edge i is the unique positive solution of

$$G_i(s) := \sum_{m \neq i} \frac{1}{1 + e^{s\ell_m}} = A.$$

If $\ell_i \geq \ell_j$, then

$$G_i(s) - G_j(s) = \frac{1}{1 + e^{s\ell_j}} - \frac{1}{1 + e^{s\ell_i}} \geq 0 \qquad (s > 0). \tag{2.7}$$

Both functions are strictly decreasing, so $h_i \geq h_j$. Strict inequality of the lengths makes (2.7) strict and gives $h_i > h_j$. Equal lengths give equal functions. □

3 A concavity reduction

If the original parent entropy is h_0, replace every edge length by h_0 times that length. By (2.3), the parent now has entropy one and each child entropy equals its original ratio to h_0. Writing ℓ for the normalized length vector, put

$$p_i = \frac{1}{1 + e^{\ell_i}}. \tag{3.1}$$

Every coordinate belongs to $(0, 1/2)$, and Lemma 2.1 fixes their sum at $A = 1/2$ or $A = 1$. A longest edge corresponds to a minimum coordinate. For a test entropy $0 < s < 1$,

$$\frac{1}{1 + e^{s\ell_i}} = \Phi_s(p_i), \qquad \Phi_s(p) = \frac{p^s}{p^s + (1 - p)^s}. \tag{3.2}$$

If i_* is an index of a minimum coordinate, the deletion entropy is at least s precisely when

$$\sum_{j \neq i_*} \Phi_s(p_j) \geq A. \tag{3.3}$$

The direction of this comparison follows from the strict decrease of the scalar entropy sum with s.

Lemma 3.1. For $0 < s < 1$, the continuous function Φ_s is increasing and strictly concave on $[0, 1/2]$, with $\Phi_s(0) = 0$ and $\Phi_s(1/2) = 1/2$.

Proof. For $0 < p < 1/2$, write $q = \Phi_s(p)$. Differentiation gives

$$\Phi_s'(p) = \frac{s\,q(1-q)}{p(1-p)} \tag{3.4}$$

and

$$\Phi_s''(p) = \frac{s\,q(1-q)}{p^2(1-p)^2}\big[s(1-2q) - (1-2p)\big]. \tag{3.5}$$

Since $p/(1-p) \in (0,1)$, raising these odds to the power s increases them; hence $p < q < 1/2$. It follows that

$$s(1-2q) < 1-2q < 1-2p.$$

Thus (3.4) is positive and (3.5) is negative. Continuity supplies the endpoint values and strict concavity on the closed interval. □

We will minimize a sum of Φ_s over coordinates with a fixed total and a lower bound. The elementary concentration principle needed for this is as follows. If $x_1, \dots, x_n \in [t, 1/2]$ have prescribed sum, a concave sum $\sum_i \Phi_s(x_i)$ is minimized by filling as many coordinates as possible to $1/2$, leaving at most one intermediate coordinate and setting the remaining coordinates equal to t. To see this directly, choose two coordinates strictly between their bounds, label them $x \le y$, and transfer a quantity u from x to y. Concavity gives

$$\Phi_s(x-u) + \Phi_s(y+u) \le \Phi_s(x) + \Phi_s(y) \tag{3.6}$$

until one of the two bounds is reached. Repeating reduces the number of interior coordinates. The resulting vector is determined up to permutation by its sum and the two bounds. Equivalently, it majorizes every vector with those constraints.

Although the coordinates in (3.1) are strictly less than $1/2$, we may minimize over the larger closed set. The values 0 and $1/2$ below are therefore auxiliary endpoints; no zero or infinite edge length is added to the theorem's hypotheses.

4 Sharp deletion bounds

4.1 Roses

Let

$$s_r = \frac{\log(2r-3)}{\log(2r-1)}, \qquad t = \min_i p_i \in \left(0, \frac{1}{2r}\right]. \tag{4.1}$$

After deleting one coordinate equal to t, the retained coordinates sum to $1/2 - t$ and are all at least t. The concentrating vector from Section 3 is

$$\left(\frac{1}{2} - (r-1)t, \underbrace{t, \dots, t}_{r-2}\right). \tag{4.2}$$

Its first coordinate lies in $[t, 1/2]$. Hence

$$\sum_{j \ne i_*} \Phi_{s_r}(p_j) \ge F_r(t) := \Phi_{s_r}\left(\frac{1}{2} - (r-1)t\right) + (r-2)\Phi_{s_r}(t). \tag{4.3}$$

The function F_r is strictly concave on $[0, 1/(2r)]$. Its endpoint values are

$$F_r(0) = \frac{1}{2}, \qquad F_r\left(\frac{1}{2r}\right) = \frac{r-1}{1 + (2r-1)^{s_r}} = \frac{1}{2}. \tag{4.4}$$

It follows that $F_r(t) \geq 1/2$, with strict inequality for $0 < t < 1/(2r)$. Equation (3.3) gives the rose bound.

If equality holds for positive lengths, then (4.3)–(4.4) force $t = 1/(2r)$. All r coordinates are at least this value and sum to $1/2$, so they are equal. Equation (3.1) then makes all normalized lengths equal to $\log(2r-1)$. Conversely, equal lengths give equality by (2.6). Together with Lemma 2.2, this proves the rose part of Theorem 1.1.

The constants s_r increase strictly with r: in

$$1 - s_r = \frac{\log((2r-1)/(2r-3))}{\log(2r-1)}, \tag{4.5}$$

the positive numerator decreases and the positive denominator increases. Their least value is consequently $s_3 = \log 3/\log 5$. This proves both rose assertions following Proposition 4.2 of [1].

4.2 Parallel-edge graphs

For B_k, set

$$s_k = \frac{\log(k-2)}{\log(k-1)}, \qquad t = \min_i p_i \in (0, 1/k], \qquad t_* = \frac{1}{2(k-1)}. \tag{4.6}$$

The retained coordinates have sum $1 - t$. For $0 \leq t \leq t_*$, their concentrating vector is

$$\left(\frac{1}{2},\, \frac{1}{2} - (k-2)t,\, \underbrace{t, \ldots, t}_{k-3}\right). \tag{4.7}$$

The second coordinate is at least t exactly on this interval. For $t_* \leq t \leq 1/k$, the concentrating vector is instead

$$\left(1 - (k-1)t,\, \underbrace{t, \ldots, t}_{k-2}\right). \tag{4.8}$$

Its first coordinate lies in $[t, 1/2]$. These vectors agree at $t = t_*$.

Thus the left side of (3.3) is bounded below by

$$F_k(t) = \begin{cases} \dfrac{1}{2} + \Phi_{s_k}\left(\dfrac{1}{2} - (k-2)t\right) + (k-3)\Phi_{s_k}(t), & 0 \leq t \leq t_*, \\ \Phi_{s_k}(1-(k-1)t) + (k-2)\Phi_{s_k}(t), & t_* \leq t \leq 1/k. \end{cases} \tag{4.9}$$

Each branch is concave. The two outer endpoint values are

$$F_k(0) = 1, \qquad F_k(1/k) = \frac{k-1}{1 + (k-1)^{s_k}} = 1. \tag{4.10}$$

To control both branches it remains to evaluate their common endpoint:

$$F_k(t_*) = \frac{1}{2} + \frac{k-2}{1 + (2k-3)^{s_k}}. \tag{4.11}$$

The numbers 1 and $2k-3$ have midpoint $k-1$. Strict concavity of x^{s_k} gives

$$\frac{1 + (2k-3)^{s_k}}{2} < (k-1)^{s_k} = k - 2. \tag{4.12}$$

Consequently $F_k(t_*) > 1$. On the first branch, concavity places the graph above the chord from $(0, 1)$ to this higher junction value. On the second, it places the graph above the chord from the junction to $(1/k, 1)$. Therefore

$$F_k(t) > 1 \quad (0 < t < 1/k), \qquad F_k(1/k) = 1. \tag{4.13}$$

No concavity across the junction is needed.

Equations (3.3) and (4.9) prove the parallel-edge bound. Equality for positive lengths forces $t = 1/k$, so all coordinates equal $1/k$ and all normalized lengths equal $\log(k-1)$. Formula (2.6) gives the converse. Lemma 2.2 completes the proof of Theorem 1.1. □

The parallel-edge constants increase strictly with k, since

$$1 - s_k = \frac{\log((k-1)/(k-2))}{\log(k-1)}$$

has a decreasing positive numerator and an increasing positive denominator. Their least value is therefore $s_4 = \log 2/\log 3$.

Figure 2 displays the two endpoint arguments in their smallest parameter cases.

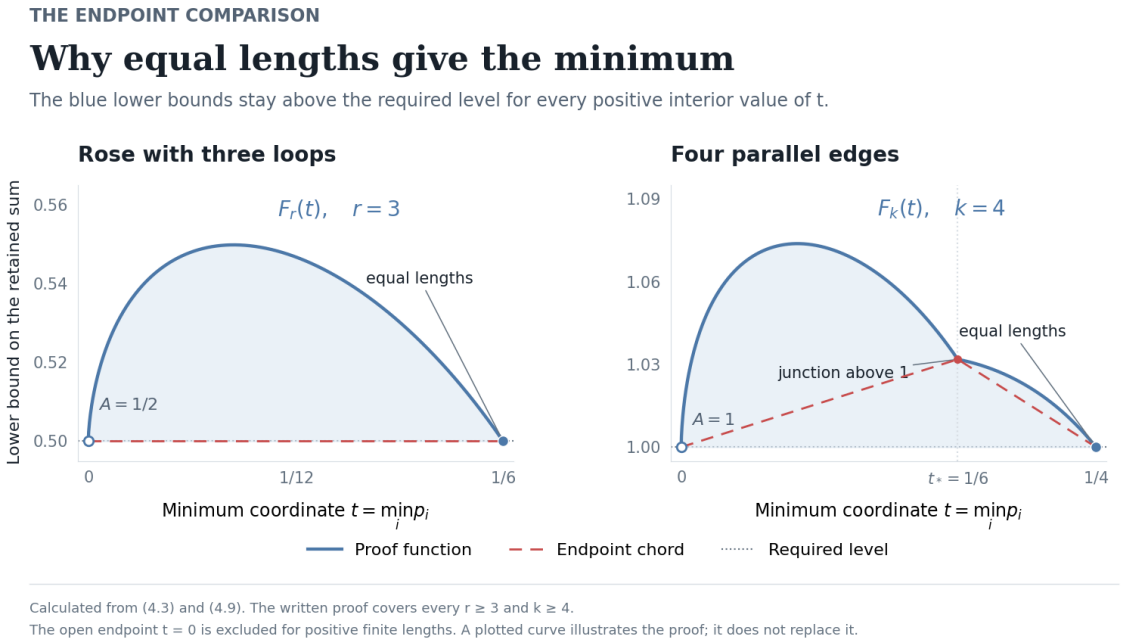


Figure 2: The proof functions F_r and F_k from (4.3) and (4.9), shown for $r = 3$ and $k = 4$ at the sharp test entropies, bound the retained transformed sums from below. The dashed chords explain why these functions exceed the required level A at every positive interior value of t; the right endpoint corresponds to equal lengths, while the open endpoint $t = 0$ is excluded for positive finite lengths. These plotted cases illustrate the analytic argument for all $r \geq 3$ and $k \geq 4$; the vertical coordinate is a transformed sum, not an entropy ratio.

5 Retaining several edges

The one-edge bounds can be iterated because each deletion leaves a graph in the same family with the inherited metric. They give exact constants for retaining any prescribed number of edges for which the remaining entropy is positive.

Corollary 5.1. Let H_j consist of the j shortest edges of the parent graph, with ties chosen arbitrarily. For a positive metric on R_r and $2 \leq j < r$,

$$\frac{h(H_j, \ell|_{H_j})}{h(R_r, \ell)} \geq \frac{\log(2j-1)}{\log(2r-1)}. \tag{5.1}$$

For a positive metric on B_k and $3 \leq j < k$,

$$\frac{h(H_j, \ell|_{H_j})}{h(B_k, \ell)} \geq \frac{\log(j-1)}{\log(k-1)}. \tag{5.2}$$

The chosen subgraph maximizes entropy among all subgraphs consisting of j full edges. Both constants are sharp, and equality holds exactly for equal parent-edge lengths.

Proof. Delete a longest remaining edge at each step. For roses, Theorem 1.1 gives

$$\frac{h(H_j)}{h(R_r)} \geq \prod_{m=j+1}^{r} \frac{\log(2m-3)}{\log(2m-1)} = \frac{\log(2j-1)}{\log(2r-1)}.$$

For parallel-edge graphs the corresponding product is

$$\prod_{m=j+1}^{k} \frac{\log(m-2)}{\log(m-1)} = \frac{\log(j-1)}{\log(k-1)}.$$

Equal lengths attain the resulting constants. If the parent lengths are not all equal, the first deletion bound is strict, so the final product is strict as well.

For optimality at fixed j, at every test entropy $s > 0$ the j shortest edges give the largest sum of the j terms $(1 + e^{s\ell_i})^{-1}$. Their scalar entropy root is consequently at least that of any other j-edge subgraph. □

Every proper subgraph formed from full edges is contained in a one-edge deletion. Since entropy is monotone under inclusion, the maximum entropy over such proper subgraphs equals the maximum over one-edge deletions. Theorem 1.1 therefore gives the sharp proper-subgraph fractions within the two stated families. Its smallest values are $\log 3/\log 5$ for roses and $\log 2/\log 3$ for parallel-edge graphs. The latter identifies the proposed four-edge extremizer within that family; comparison with other graph topologies remains open.

AI assistance

AI tools assisted the formulation, literature comparison, proof development, and preparation of this manuscript within the Mathematics Knowledge and Connections Program. AI tools were used to propose and examine arguments, compare the result with published sources, check proof steps, and revise exposition. The author directed the research selection and manuscript revisions and is responsible for the content and its final verification.

Availability of data and code

This work is theoretical and uses no experimental dataset. The source archive accompanying this preprint contains the manuscript sources, figures and Python scripts used to generate the figures. The plotted examples illustrate the analytic proof and are not used as numerical evidence for the theorems.

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